

Meeting 1022

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目錄

1 研究動機

2 文獻回顧 I

3 研究方法

4 研究資料

5 研究結果

6 參考資料

研究動機

Semiconductor Manufacturing Process

- ▶ Robust performance can be trust
- ▶ High dimension
- ▶ Find the $y = \hat{f}(x)$

研究動機

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文獻回顧 I

► Lai (2002)

1 方法

- 1 Data preprocessing
- 2 KNN-imputation
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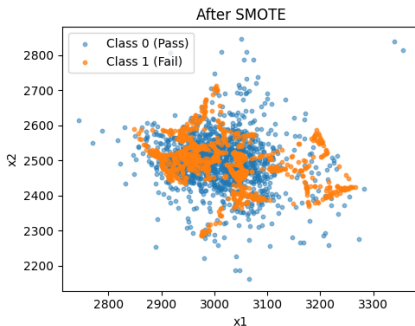
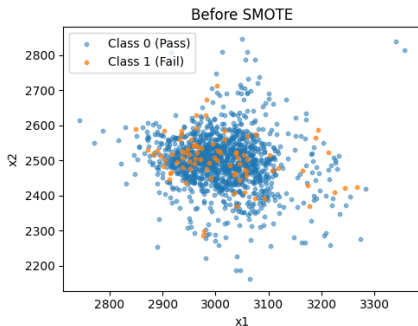
文獻回顧 I

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文獻回顧 I-方法 (SMOTE)



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2 結果

- 1 Best: ACC 86.94%, FAR 9.22%
- 2 反查 Fail 節點

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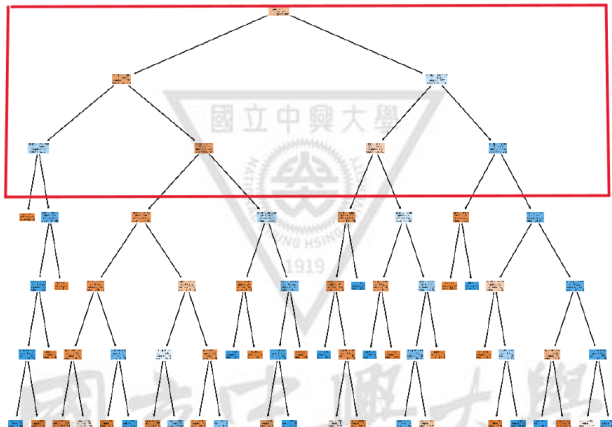
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文獻回顧 I

反查節點



文獻回顧 I

反查節點

表 4.3: 透過決策樹找出異常原因範例

Fail Case	High Concerned Factors
Data group(797)	$X(54) \leq 4.57$, $X(121) > 2.24$, $X(59) \leq 28.18$, $X(118) \leq 0.29$, $X(95) \leq -0.00$, $X(282) \leq 0.05$
Data group(929)	$X(54) \leq 4.57$, $X(121) > 2.24$, $X(59) \leq 28.18$, $X(118) \leq 0.29$, $X(95) \leq -0.00$, $X(282) \leq 0.05$
Data group(926)	$X(54) > 4.57$, $X(19) > -5464.47$, $X(110) > 15.73$, $X(153) > 0.09$, $X(31) > 8.53$, $X(236) > 0.01$
Data group(154)	$X(54) > 4.57$, $X(19) \leq -5464.47$, $X(59) > 16.34$, $X(69) \leq -0.02$, $X(244) > 0.52$
Data group(277)	$X(54) > 4.57$, $X(19) > -5464.47$, $X(110) > 15.73$, $X(153) > 0.09$, $X(31) > 8.53$, $X(236) > 0.01$

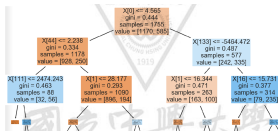


圖 4.12: 最佳化決策樹前三節點

研究方法

► 前處理

■ vote in features selection

► 機器學習模型

■ XGboost, RandomForest, DecisionTree, SVM



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研究資料

SECOM

▶ 樣本數 1567

▶ 變數 Time / x1 / x2 / ... / x590 / PassFail

■ Features: 591, Y: PassFail (binary classification)

■ Time: 2008/01/08 - 2008/12/10

■ PassFail: Pass $\rightarrow$ 0, Fail $\rightarrow$ 1 (14:1)

■ Missing values: 41951, every cols at least 1

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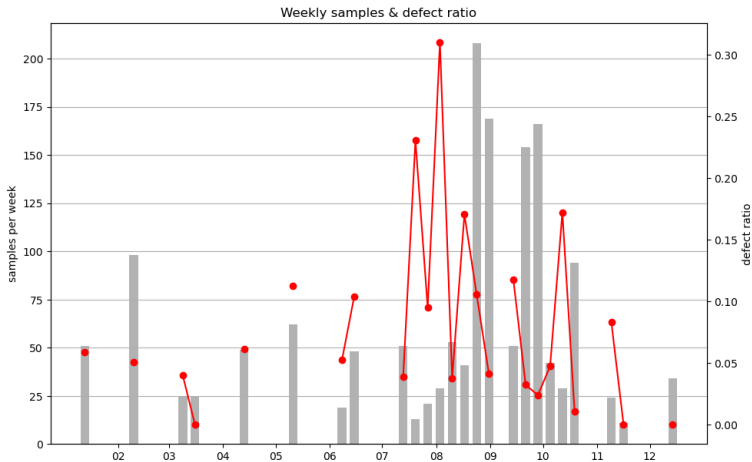
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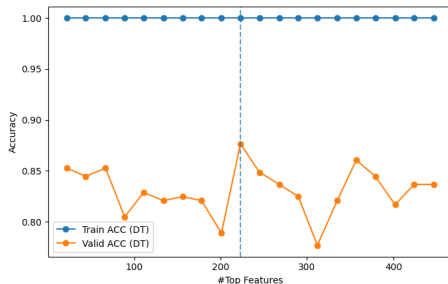


研究結果

Accuracy 0.8662
FAR 0.0922
Sensitivity 0.2857
Specificity 0.9078
GM 0.5093

Confusion Matrix

	Pred 0	Pred 1
True 0	266	27
True 1	15	6



研究結果

參考資料



Li-Fu Lai (2022)

使用 KNN 和決策樹方法對填補缺失值的製造數據進行失效預測和原因追蹤分析

The End

Questions & Comments